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MIL-DTL-14639F
16 March 2023
SUPERSEDING
MIL-DTL-14639E
14 June 1999

DETAIL SPECIFICATION

INSIGNIA, GRADE, ENLISTED PERSONNEL AND RANK, OFFICERS, METAL, GENERAL SPECIFICATION FOR

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers the general requirements of lapel buttons used by the Department of Defense and various Federal agencies (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS
MIL-STD-129 - Military Marking for Shipment and Storage
MIL-STD-1916 - DOD Preferred Methods for Acceptance of Product

Comments, suggestions, or questions on this document should be addressed to The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or email to usarmy.belvoir.hqda.mil.tioh-webmaster@army.mil. You may verify the currency of this specification using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS
MIL-F-495 - Finish, Chemical, Black for Copper Alloys

SPECIFICATION SHEETS
(See Supplement 1 for list of specification sheets)

(Copies of these documents are available online at https://quicksearch.dla.mil)

2.2.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are of these documents are those cited in the solicitation or contract.

STANDARD CHIP SETS
THE INSTITUTE OF HERALDRY
Standard Hard Enamel Color Chip Set
DoD Standard Metal Finish Chip Set

DRAWINGS THE INSTITUTE OF HERALDRY
8-3 (Series) - Insignia, Rank & Grade
B-13-12 - Attaching Devices. Heraldic, Prong and Clutch Type

(Copies of standards, drawings and Chip sets are available from The Institute of Heraldry, ATTN: Technical and Production Division, 9325 Gunston Road, Room S-108, Fort Belvoir, VA 22060-5579 or usarmy.belvoir.hqda.mi.tioh-webmaster@army.mil.)

(Figure 1 is a miniature copy of an Institute of Heraldry drawing and is for information only.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation contract.

ASTM INTERNATIONAL
ASTM B568 - Standard Test Method for Measurement of Coating Thickness by XRay Spectrometry
ASTM B487 - Metal and Oxide Coating Thickness by Microscopical Examination of a Cross Section, Measurement of
ASTM-B-86 - Zinc-Alloy Die Casting
ASTM-D-2240 - Rubber Property - Durometer Hardness
(Copies for these documents are available online at https://www.astm.org)

AMERICAN SOCIETY FOR QUALITY CONTROL
ANSI/ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes
(Copies for these documents are available online at https://acq.org)

SAE INTERNATIONAL

SAE-AMS-STD-595 - Colors Used in Government Procurement

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(Copies for these documents are available online at https://sae.org)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Tools. Is the generic name used to describe the government furnished hub or die.

3.1.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between the requirements of this specification and the specification sheets, the latter shall govern. The design of the insignia shown on the specification sheet is for illustrative purposes only. The design shall be controlled by the government furnished hub (see 3.5).

3.2 First article. When specified (see 6.2) a sample shall be subjected to first article inspection in accordance with 4.4.

3.3 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmental preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements and promotes economically advantageous life cycle costs.

3.4 Materials. Materials will bel as specified herein.

3.4.1 Copper base alloy. The copper base alloy shall be roll polished, free from pits, scale (including red oxide), dents, nicks, cracks, scratches, segregation and foreign inclusions that cannot be removed in later processing. When tested as specified in 4.5.1, the chemical composition of the copper content of the copper base alloy shall be as specified in Table I.

TABLE I. Unified Numbering System and Copper content of copper base alloys.

UNS ALLOY NUMBER	ALLOY NAME	COPPER
C210	Gilding Metal	94.0%-96.0%
C220	Bronze	89.0%-91.0%
C230	Red Brass	84.0%-86.0%
C240	Low Brass	78.5%-81.5%
C268	Yellow Brass	64.0%-68.5%
C280	Free Cutting Brass	59.0%-63.0%
C752	Nickel Silver (65-18)	63.0%-66.5% 1/

1/ When nickel silver is used as a base material, the nickel content will be not less than 18% (C752).

3.4.2 Gold.

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3.4.2.1 Gold alloy. When gold alloy is specified as a material of construction, the alloy will be of the karat quality specified on the applicable specification sheet.

3.4.2.2 Gold for plating. Gold for plating will be 24 karat.

3.4.3 Zinc aluminum alloy. The alloy used for die-casting will conform to alloy AG 40a or AC 41a of ASTM-B-86-13.

3.4.4. Nickel for plating. Nickel for plating will be the nickel plating normally used in commercial practice.

3.4.5 Rhodium Plating. Rhodium for plating will be the rhodium plating normally used in commercial practice.

3.4.6 Solder.

3.4.6.1 Hard Solder. Hard solder will be a silver-colored hard solder having a melting point of not less than 1075 °F (579.44 °C).

3.4.7 Enamel. All enamel will be lead free.

3.4.7.1 Epoxy resin (stoning epoxy). Epoxy resin will be a stoning epoxy pigmented to produce the required color, opacity or translucence, shade incorporating an ultraviolet inhibitor and have a shore D hardness of 77.

3.4.7.2 Hard enamel. Hard enamel will be a glass, vitreous type enamel fused with metallic oxides to produce the required color, opacity or translucence, and shade.

3.4.7.3 Soft enamel. Soft enamel will be a soft baking enamel pigmented to produce the required color and shade.

3.4.7.4 Baking enamel. The enamel for subdued insignia will be a flat, synthetic baking enamel pigmented to match the standard established in FED-STD-595 as follows:

COLOR	STANDARD SHIP NUMBER
Black	37038
Brown	30099
Green	34096

3.4.8 Lacquer. Lacquer will be a semi-gloss, clear, synthetic lacquer. The use of a pigmented lacquer will not be permitted.

3.5 Design. Except for Army enlisted grade insignia, the embossed design of all insignia will be an exact duplicate of the design on the Government loaned hub (see 3.8), from which the contractor's working die will be extracted. The contractor's working die will be tooled and polished to remove any dents, nicks, scratches, or other imperfections.

3.5.1 Army enlisted grade insignia. Except for the Sergeant Major of the Army and the eagle for Specialist Four grade insignia, all Army enlisted grade insignia shall conform to the details and dimensions shown on TIOH drawing B-3-83. The Sergeant Major of the

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Army grade insignia and the eagle for the mirror-like polished Specialist Four grade insignia shall be an exact replica of a Government furnished hub (see 3.8) from which the contractor's working dies shall be extracted. The base of the Specialist Four grade insignia shall conform to the details and dimensions shown on TIOH drawing B-3-83A. When the subdued Specialist Four grade insignia is manufactured as a one-piece item, the insignia shall conform to the details and dimensions shown on TIOH drawing B-3-83A. A forcer may be used to bring up the design of the eagle. The impression of the forcer will be regular and symmetrical with the design of the insignia front. The insignia back shall have no sharp or rough edges or corners.

.6 Construction.

3.6.1 Stamping, trimming, and piercing. The stamping will have a well-defined die struck edge. The insignia shall be trimmed and, when applicable, pierced to the die struck edge. All edges will be clean, smooth, and free from burrs, drag, step, and rough edges. The stamping, piercing, and trimming operations will not damage or distort the design or alter the shape of the item.

3.6.2 Die casting. When die cast method is applicable, the item will be free from porosity, blow holes, or other casting defects. Before finishing, all flash lines shall be removed, and the edges of the item will be smooth. Items that are die cast from zinc aluminum specified in 3.4.3 will be copper plated. When tested as specified in 4.5.2.2, the plating will be a minimum of 0.0003 inch thick.

3.6.3 Hard soldering and electronic fusion. Unless otherwise specified on the applicable specification sheet, all soldering will be accomplished using hard solder specified in 3.4.7.2 or by electronic fusion. Joints shall be clean, smooth, strong, and free from burned or reduced areas. There shall be no excess solder and all excess flux will be removed. When hard solder is used, the soldered parts or prongs will not separate at the joint when tested as specified in 4.5.4.1. When fused joints are used, the prong will not separate at the joint when tested as specified in 4.5.5.

3.6.4 Forcers. When forcers are permitted to bring up the design, the back of the insignia will have no sharp or rough edges. The impression of the forcer will be symmetrical with the embossed design of the insignia. Sufficient metal will be left near any trimmed area to assure clean, smooth trimming without distortion. Unless otherwise specified on the applicable specification sheet, mock forcers will not be permitted. When mock forcers are permitted, the forcer will be used only on the areas specified.

3.6.6 Attaching devices. Attaching devices will be as specified on the applicable specification sheet.

3.6.6.1 Prong (registering). Prongs will be made from nickel silver specified in 3.4.1. The prongs will be of the dimensions shown on the applicable specification sheet. They will be positioned and attached as required by the applicable specification sheet. Unless otherwise specified, the prong will be secured to the insignia by soldering or electronic fusion.

3.6.6.2 Integral prong. When prongs are die-cast as integral parts of the insignia and tested in accordance with 4.5.2.1, no evidence of breaking or cracking will occur.

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3.6.6.2 Prong and wing-type clutch. Prongs will be made from nickel silver or brass specified in 3.4.1. Unless otherwise specified on the applicable specification sheet, the prongs as shown on TIOH drawing B-13-12 (Figure 1) will be 5/16 inch ± 1/64 inch long. The prongs will be hard soldered or electronically fused in the location specified in the applicable specification sheet. Wing-type clutches as shown on TIOH drawing B-13-12 (Figure 1) will be made from any type of brass. When tested as specified in 4.5.8.1 it will not be possible to remove the clutch from the prong without first depressing the release mechanism.

3.6.6.3 Prong with flat ball-type clutch. Prongs will be made from nickel silver or brass specified in 3.4.1. Unless otherwise specified on the applicable specification sheet, the prongs as shown on TIOH drawing B-13-12 (Figure 1) shall be 5/16 inch ± 1/64 inch long. The prongs will be hard soldered or electronically fused in the location specified in the applicable specification sheet. Flat ball-type clutches as shown on TIOH drawing B-13-12 (Figure 1) will be made from any type of brass. Nickel plated flat ball clutches shall be made from any type of brass. Nickel plated flat ball clutches will be used on lapel buttons having a silver-plated finish, and red brass flat ball clutches will be used on brass or gold-plated lapel buttons. When tested as specified in 4.5.8.2 it will not be possible to remove the clutch by hand without pulling up on the ball.

3.7 Finish. Unless otherwise specified, the front and edges (surfaces visible from the front) will be finished as specified on the applicable specification sheet and will match the TIOH standard enamel color or metal finish chips as required (see 2.2.2). All brass, mirror-like polished insignia shall be lacquered. When plating is specified, the entire insignia, excluding attaching devices, will be plated. The backs of brass, gold, nickel, and rhodium plated insignia will be clean and bright. Except for Navy and Marine Corps insignia, gold plated insignia may be lacquered at the option of the manufacturer. All Navy and Marine Corps gold plated insignia will be lacquered. When an insignia is lacquered, the front surface and edges shall be lacquered using lacquer specified in 3.4.8. Lacquering of the entire insignia is optional.

3.7.1 Subdued insignia. The face and edges of all subdued insignia will be coated with enamel specified in 3.4.7.3. Prior to enameling the entire insignia will be given a black chemical finish conforming to MIL-F-495. When the subdued insignia is die-cast from zinc aluminum specified in 3.4.3, the insignia shall be copper plated prior to finishing. When tested as specified in 4.5.1, the plating will be a minimum of 0.003 inch thick. When tested as specified in 4.5.7.2, the enamel will not flake or chip.

3.7.2 Plating. All plating will be by electroplating methods. The plating will be non-porous and continuous over the entire plated surface. There will be no cut-through, shaded, peeled, or blistered plating. Plating will be smooth, fine grained, adherent, and free from pits, nodules, porosity, indications of burning, excessive edge build up, and other detrimental defects. Unless otherwise specified, the plating of attaching devices is not required.

3.7.2.1 24 karat gold plating. The gold plating will be accomplished using gold specified in 3.4.2.1. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating with 24 karat gold shall not be permitted. When tested as specified in 4.5.3, no visible chemical reaction such as evolution of gases will appear.

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3.7.2.2 22 karat gold plating. The gold plating will be accomplished using gold specified in 3.4.2.2. Prior to plating, all surfaces shall be cleaned of all matter that may affect the gold plating. The use of nickel as an undercoating shall be permitted. When tested as specified in 4.5.3, no visible chemical reaction such as evolution of gases will appear.

3.7.2.3 Nickel plating. Nickel for plating will be accomplished using nickel specified in 3.4.4. The nickel plating will be not less than 0.0003 inch thick. Testing will be standard commercial. In case of a dispute, testing will be in accordance with ASTM-B-487.

3.7.2.4 Nickel underplating. When nickel underplating is specified, nickel specified in 3.4.4 will be used and will be not less than 0.00025 inch thick. Testing will be standard commercial. In case of a dispute, testing will be in accordance with ASTM-B- 487.

3.7.2.5 Rhodium Plating. When rhodium plating is specified, rhodium specified in 3.4.5 will be used. When tested as specified in 4.5.3, no chemical reaction shall occur. When rhodium plating is required over brass, nickel underplating specified in 3.4.4 will be required.

3.7.2 Enameling. Unless otherwise specified on the applicable specification sheet, all enameling will be accomplished using epoxy specified in 3.4.6.1 or hard enamel specified in 3.4.6.2. Epoxy or enamel will be applied to be uniform in color, free from bubbles, foreign inclusions, cracking, crazing, or other defects which might affect appearance. There will be no over-running of epoxy or enamel.

3.7.2.1 Colors. Enamel or epoxy colors will match the shades of the Institute of Heraldry hard enamel color chips specified on the applicable specification sheet (see 2.1.2).

3.7.2.2 Epoxy Resin. Epoxy resin specified in 3.4.8.2 will be applied to be uniform in color, free from bubbles, foreign inclusion, or other defects which affect appearance. The epoxy will be stoned level with the dikes and polished to produce a glass like finish. No design elements will be removed during the stoning process. When tested as specified in 4.5.7.1, the epoxy will have a shore D hardness of not less than 85.

3.7.2.3 Hard enameling. Hard enamel specified in 3.4.8.1 will be charged, fired, and stoned level with the dikes. No design elements shall be removed during the stoning process. There will be no noticeable burn spots because of the firing. The hard enamel will then be polished to produce a glass like finish. The finish will be adherent and free from bubbles, pits, foreign inclusions, cracking, crazing, burned edges, or darkened enamel.

3.7.2.4 Soft enamel. Soft enamel shall be as specified in 3.4.8.4. When tested as specified in 4.5.7.3, no color will be transferred to the cotton, and the enameled surface will remain unaffected except for a slight loss of luster.

3.7.2.5 Baking enamel. When the item is to be subdued, the face and edges will be coated with enamel specified in 3.4.8.3. Prior to enameling, the entire insignia will be given a black chemical finish conforming to MIL-F-495. When tested as specified in 4.5.7.2, the baking enamel will not flake or chip.

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3.7.3 Cutting down. When necessary to remove nicks, scratches, pin holes, or other blemishes, the item will be cut down by means of grease or oil buffing. The insignia will not be cut down to the extent that any details of the design are obliterated or reduced.

3.7.4 Lacquering. The front, back, and edges of all brass, gilding metal, silver, or silver-plated items will be coated with clear lacquer specified in 3.4.9. Gold plated and Rhodium plated parts that have a matte finish or matte finish with polished highlights will be lacquered. The lacquer film will be continuous, level, adherent, and free from lint, dust, and other foreign matter. When tested as specified in 4.5.6, the tissue paper shall not adhere to the clear lacquered surface.

3.8 Government loaned property. Hubs will be loaned by the Government and will be used to make the contractor's working dies necessary for one contract or order (see 6.4).

3.9 Marking for identification. The contractor will stamp his identifying mark and the letters "GI" (GI for government procurement only) legibly and inconspicuously on the back of each item. When practical, the items will be stamped in the forced areas.

3.10 Marking for cleaning. Marking for cleaning of gold-plated items will be as follows: "To clean, rub gently with a soft, dry cloth. Do not clean with a chemically treated cloth or abrasive".

3.11 Workmanship. The finished item will be clean, well made, and will conform to the acceptable quality levels established by this specification.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Conformance inspection (see 4.4).

4.2 Inspection conditions. Unless otherwise specified, all inspection will be performed in accordance with the test conditions specified in 4.4 and 4.5.

4.3 First article inspection. Inspection and testing of the first article (see 3.2) will be made of a finished item for all provisions of this specification applicable to the end product examination and tests.

4.4 Conformance inspection. Inspection will be in accordance with the provisions set forth herein, except where otherwise indicated (see 6.3). Sampling for inspection will be in accordance with ANSI/ASQC Z1.4.

4.4.1 Testing of components. Inspection will be performed on components and materials listed in Table II for the test characteristics shown therein. Material listed in Table II may be accepted on a contractor's certificate of compliance for requirements specified in applicable paragraphs of this specification. The government reserves the right to test samples of gold (without additional charge) to verify the contractor's certification.

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TABLE II. Testing of components.

COMPONENT	CHARACTERISTIC	RQM'T 1J	TEST METHOD	RESULTS REPORTED AS
Copper base alloy	material identification	3.4.1	4.5.1	nearest 0.1%
Zinc aluminum alloy	material identification	3.4.3	standard commercial	pass/fail
Rhodium plating	material identification	3.4.5	standard commercial	pass/fail
Enamel (hard)	material identification	3.4.7.1	standard commercial	pass/fail
Enamel (soft)	material identification	3.4.7.4	standard commercial	pass/fail
Enamel (baking)	material identification	3.4.7.3 & 3.7.3.4	4.5.7.2	pass/fail
Epoxy resin	material identification	3.4.7.2	standard commercial	pass/fail
Gold for plating	material identification & karat	3.4.2.1 & 3.4.2.2	standard commercial	pass/fail
Lacquer	material identification	3.4.8	standard commercial	pass/fail
Nickel for plating	material identification	3.4.4 & 3.7.2.3	standard commercial	pass/fail
Nickel for underplating	material identification	3.4.4& 3.7.2.4	standard commercial	pass/fail
Nickel silver	material identification	3.4.1, 3.6.5.1 3.6.5.3, & 3.6.5.4	standard commercial	nearest 0.1%
Solder, hard	*material* identification	3.6.3	standard commercial	pass/fail

4.4.2 In-process inspection. Inspection will be made at any point or during any phase of the manufacturing process to determine whether operation or assemblies listed in 3.5, 3.6, and 3.7 are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material for which in-process inspection has indicated nonconformance.

4.4.3 End item inspection.

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4.4.3.1 Visual examination of insignia. Visual examination will be made using Table III. Examination shall be made at approximately 16 to 22 inches with illumination equal to average daylight and arranged so as to avoid as much reflected light as possible. Defects designated by and asterisk (*) will be classified as major when seriously affecting appearance or serviceability and minor when affecting appearance or serviceability but not seriously. The product unit for the examinations shall be one completely fabricated insignia. Any dimension which is not within the specified tolerance shall be classified as a defect.

TABLE III.
Defects.

EXAMINE	DEFECT	CLASSIFICATION	
		MAJOR (*)	MINOR
Metal color and finish	Poor match to the TIOH Standard Metal Finish Chip	X	
	Not required finish		*
	Discoloration, such as spot, stain, or speck		*
	Not clean		*
	Paired insignia not matching in color, shade and finish (when applicable		
	Background or recessed areas not finished as specified	X	
	Surface contains pits, scale, nick, scratch, crack, machine mark, pin hole, segregation, rupture, foreign matter, or other blemish		*
	Buff drag, blushed, or cloudy finish		*
Hard enamel and epoxy	Not enameled (when required)	X	
	Incorrect color or shade	X	
	Cracked, chipped, or crazed		*
	Not stoned smooth, i.e., is coarse, uneven, or has drag marks		*
	Not level with dikes, i.e., contains high or low spots		*
	Bubbles, blisters, holes, lumps, or foreign inclusions		*
	Laps, overruns, or skips		*
	Blemished, i.e., spot, stain, spatter, burned area, lack of gloss		*

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Soft/baking enamel	Not enameled or area of no enamel	X	
	ncorrect color or shade		
	Area of thin enamel	X	
	Baking enamel not flat and dull		
	Not adherent, i.e., blistered, flaking, or peeling.		*
	Foreign matter imbedde.....		*
	Discoloration		*
	Coating not continuous and level, i.e., runs, drips, or drops	X	
Design	Construction or design details obscured, altered, or not conforming to Government hub	X	
	Not dapped (when required)		
	Any significant detail not clear, altered, reduced, distorted, or missing	X	
	Any warp, twist or distortion, irregular Surface contour or outline …….		*
	Any detail struck over resulting in a double impression ……..	X	
	Any area not pierced (when required) …..	X	
Plating	Not plated (when required) …….	X	
	Not type specified …….	X	
	Plating is not smooth, continuous, or adherent ……	X	
	Foreign matter imbedded ……	X	
	Surface or edge not clean, not smooth or free from burns, roughness, drag, step, or tool marks ……		*
Workmanship and Construction	Metal marks or exposed surface (such as nick, dent, dig, gauge, or scratch) …….		*
	Die-struck edge, rim, or outline is not defined when applicable ……	X	*
	Not trimmed to the die -struck edge, when applicable ……..		*
	Pierced edge not clean and smooth, i.e., any burr, drag, cutter, or file mark ……		*
	Design or shape is damaged, disorder, or altered by piercing operation …….		*
	Forcer used when not specified or insignia forced in other than specified location …		*
	Metal torn or broken out by use of force ….		*
	Any component missing, bent, twisted, broken, or malformed ……		
	Soldered does not completely unite parts i.e., any perceptible opening between components parts ……		*
	Any area burned or reduced in soldering ….		*
	Soldered joint not clear or smooth, excess		

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Attaching device	solder or excess flux not removed …….	X	
	Back not flat or smooth when required …….	X	
	Superimposed design not securely riveted to insignia ……..		*
	Not specified type size, or material …….	X	
	Not attached in specified manner …….	X	
	Any component missing …….	X	
	Defective, i.e., any part damaged or Malformed affecting use ……		*
	Defective, does not operate as intend.....		*
Identification	Notpositioned as specified	X	
	Clutch does not engage prong or is loose fit.		
	Clutch pierced through by prong	X	
	Prong loose	X	
	Missing, incorrect, illegible, misspelled, not accomplished as specified, or not placed as required		*
	Not lacquered when required		
Lacquer	Areas of no lacquer.....	X	
	Foreign matter embedded in finish, i.e., lint, dust, etc.....		
	Hazy, rainbow effect, cloudy, or powdering		
	Not smooth, continuous, or adherent, i.e., flaking, blistering, peeling, rippling, or has run, sag, or drops		*
	Not set to touch, i.e., tacky when pressure is applied to coating	X	
Quality of Unplated metal	Surface pitted, porous, crystalline, potted, opened grained, or otherwise impaired……..	X	
		X	

4.4.3.2 End item testing. Testing of the completely fabricated item shall be performed in accordance with Table IV for the characteristics shown therein. The sample unit shall be one completely fabricated badge, badge clasp, bar, or pendant. The sample unit for the test for hard soldered or fused joints shall be one finished item. All items shall be tested, and one defect shall be scored regardless of how many prongs on that item failed. Individual tests are detailed in 4.5.

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TABLE IV. **Testing of end item.**

CHARACTERISTIC	RQM'T	TEST METHOD	RESULTS REPORTED AS
Test for copper content	3.4.1	4.5.1	pass/fail
Tests for plating			
Nickel plating	3.7.2.3	standard commercial	pass/fail
Nickel underplating not permitted with 24 karat gold plating	3.7.2.1	standard commercial	pass/fail
Nickel underplating with 22 karat gold plating	3.7.2.2	standard commercial	pass/fail
Copper plating	3.7.1	standard commercial	pass/fail
Acid test:	3.7.2.1	4.5.3	pass/fail
24 karat gold plating 22 karat gold plating	3.7.2.2	4.5.3	pass/fail
Rhodium plating	3.7.2.5	4.5.3	pass/fail
Test for lacquer: Tests for joints:	3.7.5	4.5.6	pass/fail
Hard soldered joints	3.6.3	4.5.4	pass/fail
Electronically fused joints	3.6.3	4.5.5	pass/fail
Tests for attaching devices:			
Integral prong	3.6.5.2	4.5.2.1	pass/fail
Prong with wing-type clutch	3.6.5.3	4.5.8.2	pass/fail
Prong with flat ball clutch	3.6.5.4	4.5.8.3	pass/fail
Tests for enamels:			
Epoxy resin	3.7.3.3	4.5.7.1	pass/fail
Baking enamel	3.7.3.4	4.5.7.2	pass/fail
Soft enamel	3.7.3.5	4.5.7.3	pass/fail

4.4.3.3 Examination of Captain (Navy) or Colonel insignia for pairing. Any Captain (Navy) or Colonel insignia mismatched shall be classified as a defect. The sample unit for this examination shall be one pair of Captain (Navy) or Colonel insignia.

4.4.3.4 Examination of paired insignia for compatible finish. Any paired insignia having finish and/or color which is not closely matched to each other shall be classified as a defect. The sample unit for this examination shall be one pair of insignia mounted on a card.

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4.5 End item tests.

4.5.1 Copper content test for copper base alloy. Copper content shall be determined by standard commercial testing. Results shall be evaluated to determine compliance with the requirements specified in 3.4.1.

4.5.2 Test of Die casting.

4.5.2.1 Test for integral prongs. Insignia to be tested shall be placed in a jig which allows no movement of the insignia and positions prongs at a 30-degree angle above the horizontal. A vertical 1O pound load shall be applied to the top third of each prong. The prongs shall be examined to determine compliance with 3.6.5.2.

4.5.2.2 Direct micro thickness test for copper plating. Copper plating thickness shall be determined by standard commercial testing. Results shall be evaluated to determine compliance with 3.6.2.

4.5.3 Acid test for gold and rhodium plating. If the test insignia has been lacquered, the lacquer shall be completely removed prior to testing with an organic solvent. The test acid shall be applied as follows: Place a drop of acid of not less than 1/16 inch in diameter on three (3) different spots on the plated surface allowing the drops to remain for not less than one (1) minute during which time the surface of the item shall be inspected to determine compliance with the requirements in 3.4.2.1 and 3.4.2.2. No visible chemical reactions shall occur. Acid drops to be placed on flat surfaces on the obverse face when possible. A minimum of two spots must withstand the acid test. The test acid shall be applied at room temperature (60 to 80° Fahrenheit/15.6 to 26.7° Celsius) and shall consist of a solution containing 50% by volume of chemically pure nitric acid (specific gravity 1.42) and an equal volume of distilled water.

4.5.4 Test for hard soldered joints.

4.5.4.1 Hard solder. Hard soldered items shall be placed in an oven maintained at 1075" Fahrenheit ± 5° Fahrenheit (561.7° Celsius ± 2.78° Celsius) for 15 minutes. While at this temperature the item shall be lifted by the attaching device and shall be inspected to determine compliance with the requirements specified in 3.6.3.

4.5.5 Test for electronically fused joints. Items to be tested shall be anchored on a horizontal surface. Each prong shall be grasped at least 1/3 its length above the fused joint and bent through an angle of 90 degrees (45 degrees to each side of the vertical) until the prong breaks. The fused joint shall then be examined to determine compliance with 3.6.3. A bending tool in the form of a 45-degree template may be used for this test provided the prong is gripped at least 1/3 its length above the fused joint.

4.5.6 Test for lacquer. At room temperature (60 to 80° Fahrenheit/15.6 to 26.7° Celsius), press a piece of tissue paper against the lacquered surface for 15 seconds, using any pressure capable of being exerted between thumb and two fingers, after which the pressure shall be released, and the item inspected to determine compliance with 3.10.

4.5.7 Test tor enamels.

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4.5.7.1 Test for shore hardness of epoxy resin. Shore hardness shall be determined in accordance with ASTM-D-2240. Results shall be evaluated to determine compliance with the requirements of 3.7.3.3.

4.5.7.2 Test for baking enamel. Draw a knife edge across the face side of the specimen making four parallel scores 1/16 inch apart. Repeat this process drawing the knife edge this time perpendicular to the previously scored lines, making small blocks in the finish. The finish shall then be examined to determine compliance with 3.7.1.

4.5.7.3 Acetone test for soft enamel. The enameled surface shall be wiped five times with a piece of cotton saturated with acetone. The cotton and enamel shall then be examined to determine compliance with 3.4.7.4.

4.5.8 Test for attaching device.

4.5.8.1 Test for integral prongs. Insignia to be tested will be placed in a jig which allows no movement of the insignia and positions prongs on a 30-degree angle above the horizontal. A vertical 10-pound load shall be applied to the top third position of each prong. The prongs will be examined to determine compliance with 3.6.5.2.

4.5.8.2 Test for prong and wing-type clutch. The clutches shall be removed and replaced ten times from the prong using the clutch release mechanism. An attempt shall then be made to remove the clutch by hand without first depressing the release mechanism. An inspection shall be made at this time of the clutch and prong to determine compliance with the requirements specified in 3.6.5.3.

4.5.8.3 Test for prong and flat ball clutch. The clutches shall be removed and replaced from the prong by manually pulling up on the ball of the crutch ten times. An attempt shall then be made to remove the clutch by hand without first grasping the ball. An inspection shall be made at this time of the prong with flat ball clutch ta determine compliance with the requirements specified in 3.6.5.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. Insignia covered by this specification are intended to be worn by officers of the Army, Navy, Marine Corps, and Air Force and by enlisted personnel of the Army and Air Force on designated uniforms to indicate the rank or grade of the wearer.

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6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.
b. The title, number, and date of the applicable specification sheet (see 1.2 and 2.1).
c. When a first article is not required (see 3.2).
d. Packaging requirements (see 5.1).
e. When special marking is required (see 3.10).

6.3 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first production sample or a standard production item from the contractor's current inventory as specified in 4.4. The first article should consist of one completed item. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or tests, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Government-loaned property. The contracting officer should arrange to loan the property listed in 3.8.

6.5 Subject term (key word) listing.

Acetone Enamel,
baking Enamel,
hard Epoxy
Gold Lacquering
Lead
Nickel
Silver Nitric acid
Plating
Prong, integral
Prong, registering
Rhodium Soldering
Zinc aluminum

6.6 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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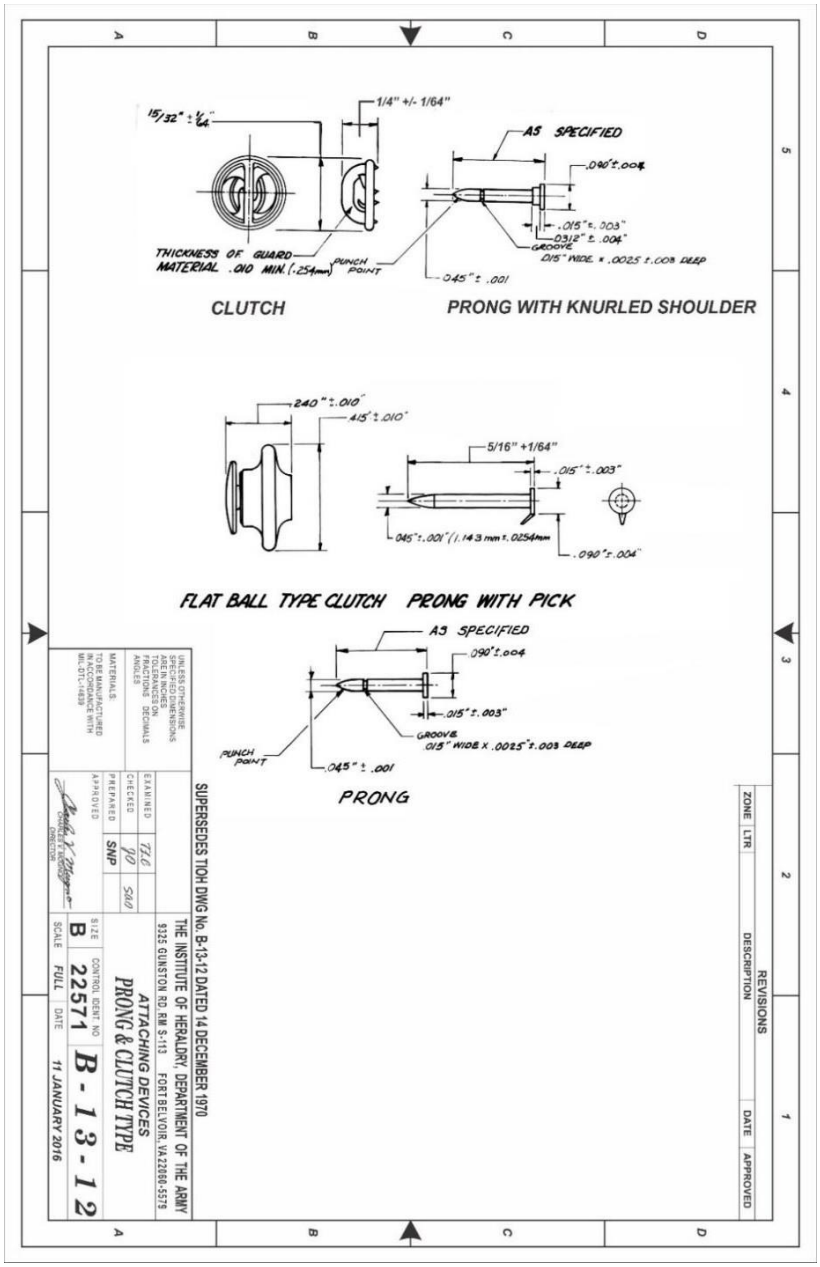


FIGURE 1. **Attaching devices heraldic, prong and clutch type B-13-12**

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Custodians:

Army - IH
Navy - NU
Air Force - 11

Preparing activity:

Army - IH

Review activities:

Navy - MC, CG1
DLA - CT

(Project No. 8455-2023-072)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the DLA ASSIST Online database at https://assist.dla.mil.

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